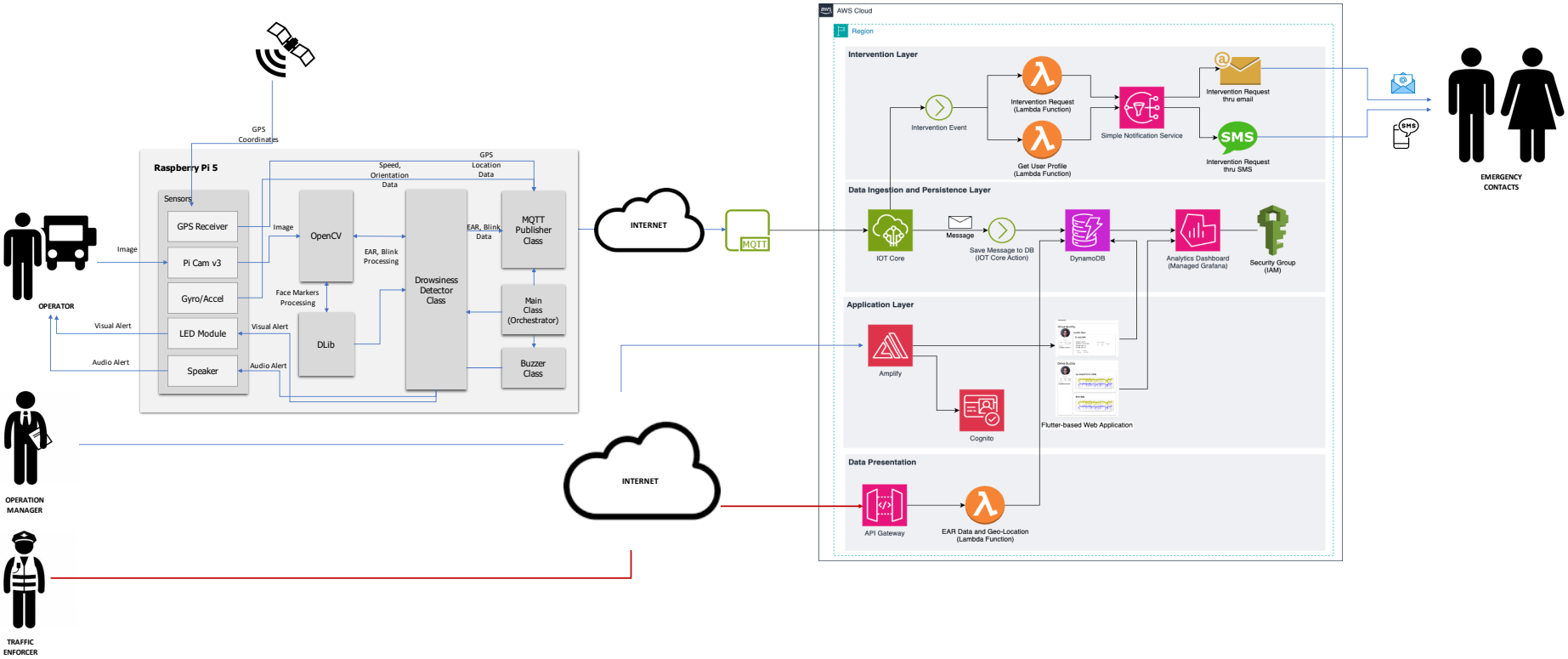
GPS Coordinates
Raspberry Pi 5
Sensors
GPS Receiver
Pi Cam v3
Gyro/Accel
LED Module
Speaker
Image
Visual Alert
Audio Alert
OPERATOR
OPERATION MANAGER
TRAFFIC ENFORCER
OpenCV
DLib
Face Markers Processing
EAR, Blink Processing
Speed, Orientation Data
GPS Location Data
Drowsiness Detector Class
EAR, Blink Data
MQTT Publisher Class
Main Class (Orchestrator)
Buzzer Class
INTERNET
MQTT
AWS Cloud
Region
Intervention Layer
Intervention Event
Intervention Request (Lambda Function)
Get User Profile (Lambda Function)
Simple Notification Service
Intervention Request thru email
Intervention Request thru SMS
Data Ingestion and Persistence Layer
IOT Core
Message
Save Message to DB (IOT Core Action)
DynamoDB
Analytics Dashboard (Managed Grafana)
Security Group (IAM)
Application Layer
Amplify
Cognito
Flutter-based Web Application
Data Presentation
API Gateway
EAR Data and Geo-Location (Lambda Function)
EMERGENCY CONTACTS